

TABLE VII
(CONTINUED)

	1. 1970– 1980 extensive margin	2. 1980– 1990 extensive margin	3. 1990– 1998 extensive margin	4. 1970– 1998 extensive margin	5. 1980– 1998 extensive margin	6. 1980– 1998 intensive margin	7. 1980– 1998 extensive + intensive
D. 10× annual changes in college-equivalent share of employment in percentage points, predicted by impact of computerization on task input (panel B)							
	NIPA computer investment measure				CPS computer use measure		
Nonroutine tasks	0.40	0.65	1.10	0.69	1.41	0.40	1.81
Routine tasks	0.29	0.48	0.81	0.51	0.80	1.04	1.84
All tasks	0.70	1.12	1.91	1.19	2.21	1.44	3.65
E. Estimated log demand shifts for college-equivalent/noncollege-equivalent labor 1970–1998 (100 × annual log changes)							
Using constant-elasticity of substitution model to estimate changes in college demand							
$\sigma = 0.0$	4.99	2.53	2.25	3.33			2.41
$\sigma = 1.4$	3.95	4.65	2.76	3.86			3.81
$\sigma = 2.0$	3.50	5.56	2.98	4.09			4.41
Using task model to predict changes in college demand							
Total task Δ (panel C)	1.23	1.29	1.43	1.31	1.56	−0.06	1.51
Predicted by computerization (panel D)	0.64	0.70	0.98	0.76	1.39	0.91	2.29

Panels C and D: Implied employment share changes for college-equivalent labor (in percentage points) are calculated as the inner product of observed or predicted changes in task input from panels A and B and the coefficient vector from a fixed coefficient model of educational input. For extensive margin task shifts, this coefficient vector is estimated from a regression of college equivalent employment (in FTEs) in 140 consistent CIC industries on the five DOT measures of industry task input (in centiles) and a constant using the 1980 MORG sample. For intensive margin task shifts, the coefficient vector is estimated from a regression of college equivalent employment in 470 COC occupations on the five DOT measures of occupational task input (in centiles) and a constant using the 1984 CPS sample. College-equivalents labor is defined as all workers with college or greater education plus half of those with some college.

Panel E: Fixed coefficients log relative demand shifts are calculated as the change in the log ratio of college-equivalent/noncollege-equivalent employment using the initial (1970, 1980, or 1990) college-equivalent/equivalent employment share in full-time equivalents and the implied percentage point change in this share from Panels C and D.

Constant Elasticity of Substitution (CES) implied relative demand shifts for college-equivalent labor are calculated following Autor, Katz, and Krueger [1998] using a CES aggregate production function with two inputs, college and high school equivalents, and an elasticity of substitution denoted by σ . See Table II of Autor, Katz, and Krueger for details.